

NET731 Practical

Week 2

Network Design

1.creating Virtual Machine

The screenshot displays the Microsoft Azure portal interface. At the top, the header bar includes the Microsoft Azure logo, a search bar, and a Copilot button. Below the header, the main content area shows the deployment details for a resource named "CreateVm-canonical.ubuntu-24_04-lts-server-20260318160418". The deployment is marked as "Complete" with a green checkmark. A sidebar on the left lists navigation options: Overview, Inputs, Outputs, and Template. The main content area includes a "Deployment details" table with columns for Resource, Type, Status, and Operation details. The table lists five resources, all with a status of "OK". Below the table, there is a "Next steps" section and a "Give feedback" link.

Home

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

CreateVm-canonical.ubuntu-24_04-lts-server-20260318160418 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

✓ Your deployment is complete

Deployment name: CreateVm-canonical.ubuntu-24_04-lts-server-2... Start time: 3/18/2026, 4:11:41 PM
Subscription: Azure for Students Correlation ID: 263b0144-e4da-4c44-b661-d46ade9bd598
Resource group: CTU-Student-RG

Deployment details

Resource	Type	Status	Operation details
✓ CtuStudent-Vnet	Microsoft.Compute/virtualMachines	OK	Operation details
✓ ctustudent-vnet288_z1	Microsoft.Network/networkInterfaces	OK	Operation details
✓ CtuStudent-Vnet-ip	Microsoft.Network/publicIPAddresses	OK	Operation details
✓ CtuStudent-Vnet-nsg	Microsoft.Network/networkSecurityGroups	OK	Operation details
✓ CtuStudent-Vnet-vnet	Microsoft.Network/virtualNetworks	OK	Operation details

Next steps

Give feedback

[Tell us about your experience with deployment](#)

Virtual Machine has been created.

2. Configuring Subnets and address space

Home > CtuStudent-Vnet-vnet

CtuStudent-Vnet-vnet | Subnets

Virtual network

Search

+ Subnet Refresh Manage users Delete Export to CSV

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

☐	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table		
☐	default	10.0.0.0/24	-	250	-	-	-		
☐	DBSubnet	10.0.1.0/24	-	251	-	-	-		

Created and Configured Database tier:

Name: DBSubnet

Subnet address range: 10.0.1.0/24

Home > CtuStudent-Vnet-vnet

CtuStudent-Vnet-vnet | Subnets

Virtual network

Search

+ Subnet Refresh Manage users Delete Export to CSV

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

☐	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table		
☐	default	10.0.0.0/24	-	250	-	-	-		
☐	DBSubnet	10.0.1.0/24	-	251	-	-	-		
☐	AdminSubnet	10.0.2.0/24	-	251	-	-	-		

Created and Configured Administration tier

Name: AdminSubnet

Subnet Address range: 10.0.0.0/24

3.Attatching Network Security Groups

Home > Network foundation | Network security groups > CtuStudent-Vnet-nsg

CtuStudent-Vnet-nsg | Subnets

Network security group

Search

+ Associate

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

- Inbound security rules
- Outbound security rules
- Network interfaces
- Subnets**

Search subnets

Name	Address range	Virtual network
AdminSubnet	10.0.2.0/24	CtuStudent-Vnet-vnet
DBSubnet	10.0.1.0/24	CtuStudent-Vnet-vnet
default	10.0.0.0/24	CtuStudent-Vnet-vnet

Give feedback

Attached Network Security Groups on both the subnets created.

References

asudbring (2025). Quickstart: Create an Azure Virtual Network. [online] Microsoft.com. Available at: <https://learn.microsoft.com/en-us/azure/virtual-network/quickstart-create-virtual-network?tabs=portal>.

asudbring (2025). Add, change, or delete a subnet. [online] Microsoft.com. Available at: <https://learn.microsoft.com/en-us/azure/virtual-network/virtual-network-manage-subnet?tabs=azure-portal>.

asudbring (2024). Tutorial: Filter network traffic with a network security group (NSG). [online] Microsoft.com. Available at: <https://learn.microsoft.com/en-us/azure/virtual-network/tutorial-filter-network-traffic?tabs=portal>.